

REPORT OF THE

PORTFOLIO COMMITTEE ON HIGHER AND TERTIARY EDUCATION, INNOVATION, SCIENCE AND TECHNOLOGY DEVELOPMENT

ON

THE PROJECTS BEING CARRIED OUT BY VERIFY ENGINEERING AND NATIONAL BIOTECHNOLOGY AUTHORITY.
28 JUNE TO 04 JULY 2024

FIRST SESSION - TENTH PARLIAMENT

JULY 2024

Presented to Parliament July 2024

S. C. 23,2024

ORDERED IN TERMS OF STANDING ORDER No. 18:

i. At the commencement of every session, there shall be as many committees to be designated according to government portfolios as the Standing Rules and Orders Committee may deem fit.

ii. It shall be the function of such committees to examine expenditure administration and policy of government departments and other matters falling under their jurisdictions as Parliament may, by resolution determine.

iii. The members of such committees shall be appointed by the Standing Rules and Orders Committee, from one or both Houses of Parliament, and such appointments shall take into account the expressed interests or expertise of the Members and Senators and the political and gender composition of Parliament.

iv. Each select committee shall be known by the portfolio determined for it by the Standing Rules and Orders Committee.

TERMS OF REFERENCE OF PORTFOLIO COMMITTEES S.O 20

Subject to these Standing Orders a Portfolio Committee shall:

i. Consider and deal with all Bills and Statutory Instruments or other matters which are referred to it by or under a resolution of the House or by the Speaker;

ii. Consider or deal with an appropriation or money bill or any aspect of an appropriation or money bill referred to it by these Standing Orders or by or under resolution of this House;

iii. Monitor, investigate, inquire into and make recommendations relating to any aspect of the legislative programme, budget, rationalization, policy formulation or any other matter it may consider relevant of the government department falling within the category of affairs assigned to it, and may for that purpose consult and liaise with such a department;

iv. Consider or deal with all international treaties, conventions and agreements relevant to it, which are from time to time negotiated, entered into or agreed upon.

HON SPEAKERS`ANNOUNCEMENT

On Thursday, 12 October 2023, the Speaker announced that the Committee on Standing Rules and Orders nominated the following members to serve on the Portfolio Committee on Ministry of Higher and Tertiary Education, Innovation, Science, and Technology Development:

1. Hon. Gumede M.
2. Hon. Chakukura A.
3. Hon. Chaimvura N.
4. Hon. Ganyiwa B.
5. Hon. Jonga W.
6. Hon. Karenyi L.
7. Hon. Machangu P.
8. Hon Makope M.
9. Hon. Makuvire. J.
10. Hon. Murwira T.
11. Hon. Mutimbanyoka K.
12. Hon. Ngadziore T.
13. Hon. Nyandoro B.
14. Hon. Sibanda O.
15. Hon. Sibanda M. C.
16. Hon. Zhanda W.

Hon Gumede M. to be the Chairperson

1.0 INTRODUCTION

The Portfolio Committee on Higher and Tertiary Education, Innovation Science and Technology Development conducted an inquiry into the operations of Verify Engineering and National Biotechnology Authority projects from 28th of June to 4th of July 2024. Both Verify Engineering and National Biotechnology Authority being parastatals that fall within the Ministry of Higher and Tertiary Education Science and Technology Development play a pivotal role in achieving the vision of the Ministry of a heritage based higher Education, Science and Technology Development for a competitive industrialized and Modernized Zimbabwe by 2030. Therefore the above mentioned Committee as part of its oversight role resolved to do an inquiry of the above mentioned entities in order to evaluate the various projects being carried out by both entities.

2.0 OBJECTIVES

1. To familiarize with the operations of Verify Engineering and National Biotechnology Authority;

2. To propose measures to capacitate both Verify Engineering and National Biotechnology Authority so as to enhance their performance.

3.0 METHODOLOGY

In conducting the inquiry, the Committee adopted established methodologies for investigations, ensuring the highest standards of transparency. This commitment to transparency not only bolstered the integrity of the process but also significantly enhanced the credibility and reliability of the Committee's report.

3.1 ORAL EVIDENCE SESSIONS

The Portfolio Committee initiated a series of oral evidence sessions with key stakeholders to gather insights essential for their investigation. The first session took place on February 13, 2024, when the Committee welcomed representatives from the Ministry of Higher and Tertiary Education, Innovation, Science and Technology Development (MHTESTD). During this session, the Permanent Secretary, shared valuable Permanent Secretary, shared valuable perspectives on the Ministry's operations and the various entities under its jurisdiction. He highlighted several

projects currently underway, including initiatives led by Verify Engineering and National Biotechnology Authority (NBA). Following this informative session, the Committee invited Verify Engineering (VE) on May 7, 2024. The CEO, presented the detailed briefing on company's mandate and the projects it was undertaking, including significant developments in Mutare Feruka and Mkwasine in Chiredzi. The Committee further engaged with the National Biotechnology Authority on May 14, 2024, where Dr. Savadye provided an in-depth overview of the NBA's role within the Ministry, outlining various initiatives being implemented throughout the country.

3.2 VERIFICATION VISITS

Building on the insights gained from the oral evidence sessions, the Portfolio Committee embarked on a series of verification visits to assess the projects and operations of both the National Biotechnology Authority and Verify Engineering. On June 29, 2024, the Committee visited the Verify Engineering projects in Mutare Feruka, followed by a trip to the National Biotechnology Authority's Mapfuura project in Masvingo Rutenga on June 30, 2024. The Committee then explored the Mkwasine project, focusing on the innovative coal-to-liquid fertilizer initiative on July 1, 2024.The following day, the Committee assessed the National Biotechnology Regulatory offices at the Beitbridge Border Post and on July 3, 2024, the Committee visited the University of Zimbabwe Farms to examine various projects and initiatives led by the National Biotechnology Authority. Finally, on July 4, 2024 the Committee travelled to Muzarabani to inspect the Masawu projects, further solidifying their commitment to a comprehensive evaluation of the ongoing initiatives.

4.0FINDINGS

4.1VERIFY ENGINEERING (VE)

4.1.1 BACKGROUND AND THE UNPACKING OF THE PROJECTS.

During its visit to Verify Engineering's Feruka projects in Mutare, the Portfolio Committee engaged with the senior management and board members revealing several key findings. The Committee noted that Verify Engineering, was established in 2005, and initially focused on researching alternative liquid fuel sources from local resources like coal and coal bed methane

gas. A significant finding was that the company's expanded mandate in 2019 to include "heritage-based research, innovation and development," which aims for industrialization through heritage-based science and technology by 2030. The Committee recognised that Verify Engineering's focus on innovation enhances its technological capabilities, aligning with national goals for economic growth and job creation. Additionally, the findings highlighted opportunities for community engagement and local partnerships, indicating the company's potential to promote sustainable practices and contribute to regional industrialization. Overall, the Committee found out that Verify Engineering is well-positioned to drive energy solutions and broader socio-economic development, reinforcing its commitment to innovation and sustainability.

4.1.2 OVERALL PROJECTS BEING CARRIED BY VERIFY ENGINEERING

The presentation at Feruka highlighted key findings for the Committee regarding Verify Engineering (Pvt) Ltd and its strategic initiatives, which align with the objectives set by the Ministry of Higher and Tertiary Education, Innovation, Science and Technology Development (MHTESTD). The Committee found Verify Engineering has established a strong presence in the production and distribution of medical and industrial gases across Zimbabwe and the region, playing a vital role in meeting industrial needs and supporting national economic growth and job creation. Additionally, the Committee noted that the company is actively engaged in advanced research and Development to drive industrialisation, with key focus areas including lithium battery development, waste management, hydrogen production and catalysis. These findings emphasize Verify Engineering's commitment to leverage on emerging technologies and sustainable practices, balancing economic growth with environmental sustainability.

The Board Chair outlined Verify Engineering's specialization in the designing, construction, procurement, and commissioning of manufacturing plants for chemical processing and food processing, as well as energy power plants. To enhance operational capacity, the company has established several Strategic Business Units (SBUs), including VE Gases for gas solutions, VE Energy for renewable projects and VE Technical Services for engineering and construction.

Additionally, the VE Technology and Innovation Support Centre focuses on high-end research and development to promote industrialisation. However, the Board Chair noted significant challenges, the substantial investment required for these initiatives and a lack of funding to bring them to bankable stage. This presents an opportunity for the Committee to advocate for funding solutions, such as joint ventures or public- private partnerships. Overall these findings positions Verify Engineering as a pivotal player in advancing industrialisation and sustainable development in Zimbabwe, highlighting the importance of innovation and strategic partnerships in achieving national goals.

4.1.3 PROJECTS REQUIRING TECHNICAL AND INVESTMENT (CTL, IRON & STEEL, CTF)

The Committee received on site presentations from Verify Engineering, which highlighted critical findings regarding the company's ongoing and proposed projects, particularly significant funding challenges that impede their initiation. The Committee noted that several mineral-based projects, including the Coal-to-Liquid Fuels and Chemical (CTX) initiative (costing between US$ 5 billion) and the Coal-to-Fertilizers and Chemical (CTF) project (estimated at US$51 million to US$ 950 million), have remained unfunded for years. Other significant projects, such as the Iron and Steel Production Plant and a Coal Thermal Power Plant, require substantial investments, further underscoring the funding gap. The Committee recognized that this lack of funding is severely hindering progress and limiting potential economic benefits for Zimbabwe.

Verify Engineering suggested that funding could come from various sources, including foreign direct investment, public- private partnership and joint ventures. The Committee acknowledged the importance of addressing the funding gap to support these initiatives, which could significantly contribute to the country's economic growth and resource utilization. Overall, these findings emphasize the urgent need for collaborative efforts to secure funding and realize these strategic projects, reinforcing the critical role of innovation and investment in Zimbabwe's industrialization agenda.

4.1.4 VERIFY ENGINEERING MKWASINE

During the Committee's visit to Verify Engineering, at Mkwasine presentation on site outlined the company's significant potential in fostering industrialisation through its innovative coal-to-liquid fertilizer project in Chiredzi. The Committee found out that this initiative emphasizes beneficiation and effectively harness local resources. Key findings included a comprehensive feasibility study on coal gasification technology, essential for producing synthesis used in creating nitrogenous fertilisers via the Bosch-Haber Process. Verify Engineering's expertise in infrastructure development ensures efficient operation and production targets for the plant. The Committee also recognised the economic implications of local fertiliser production, which aims to reduce Zimbabwe's reliance on imported fertilizers, currently at 90% of demand, promising substantial cost savings and enhancing local agriculture's viability. Moreover, the project prioritizes environmental sustainability and benefits from collaboration with international partners like the Magco Consortium Group of Companies, which adds resources and expertise. Overall, the Committee concluded that Verify Engineering is laying the groundwork for economic growth and sustainability in the coal-to-liquid fertilizer sector, establishing a robust foundation for regional industrialisation.

4.2.1 NATIONAL BIOTECHNOLOGY AUTHORITY

During its visit to the National Biotechnology Authority (NBA) projects in Rutenga, the Portfolio Committee met with the Board and senior management, uncovering several key findings. The NBA, established under the National Biotechnology Authority of Zimbabwe Act of 2006, is an autonomous research and development institution aimed at transforming Zimbabwe from a raw material-based to a knowledge-based economy through the application of biotechnologies in agriculture, medicine, energy and environmental sectors. Dr. Savadye outlined the NBA's 19 primary roles, emphasizing its key functions, which include regulating biotechnology goods and services, developing long term biotechnology policy, promoting research and innovation, leading genome analysis, reviewing biotechnology project proposals and ensuring national biosafety and biosecurity. These findings underscore the NBA's critical role in advancing Zimbabwe's economic transformation through biotechnology.

4.2.2 NATIONAL BIOTECHNOLOGY MAPFURA AND MASAU PLANTS

The Committee found out that the National Biotechnology Authority (NBA) is instrumental in promoting industrialisation through its flagship projects: the Mapfura processing plant in Mwenezi and Masau project in Muzarabani. Both projects aim to convert locally abundant fruits into value-added products, such as wine and jams, creating direct employment opportunities and fostering local entrepreneurship. The NBA highlighted that these initiatives significantly contribute to rural industrialization and enhance economic benefits for local communities. Furthermore, ongoing research and development efforts are focused on improving processing techniques and diversifying product offering for both local consumption and export. The Committee emphasised the importance of sustainability of these projects, noting their commitment to environmentally friendly practices and community engagement, which promote responsible resource management. Overall, the Committee noted that these initiatives exemplify how biotechnology can drive rural development and empower communities, showcasing their potential to improve local economies and support sustainable livelihoods.

4.2.3 NATIONAL BIOTECHNOLOGY ACTIVITIES AT BEITBRIDGE BORDER POST

The Committee noted that NBA plays a vital role at the Beitbridge Border Post, a key entry point for trade between Zimbabwe and South Africa. During the visits the CEO emphasized on the NBA's regulatory responsibilities, particularly in streaming import and export processes for agricultural products to ensure compliance with national compliance with national and international biotechnology standards. A key finding was the NBA's implementation of biosecurity measures to protect local agriculture from pests and diseases associated with imported goods. The Committee also discovered that the NBA conducts thorough inspections of agricultural products and facilitates the safe movement of biotechnology derived goods. Additionally regulations and adopting best practices for sustainability and productivity.

The Committee recognised that the NBA actively promotes research and innovation in biotechnology at Beitbridge, collaborating with various stakeholders to develop solutions that enhance agricultural resilience and food security. These findings align with the Committee's

objectives during the familiarization visits, highlighting the NBA's capacity to facilitate industrialisation through beneficiation. Overall, the Committee found out that NBA's focus on innovation and collaboration is essential for fostering a sustainable agricultural framework capable of meeting market demands while addressing environmental concerns, thereby enhancing the competitiveness of Zimbabwe's agricultural sector in cross-border trade.

4.2.4 NATIONAL BIOTECHNOLOGY AND THE UNIVERSITY OF ZIMBABWE BIOBANK AND COLLABORATION BETWEEN NBA AND UZ FARM

The Committee findings reveal that the NBA plays a vital role in the UZ Biobank of the University of Zimbabwe, which collects and preserves biological samples for health and biotechnology research. NBA, emphasized that the biobank facilitates studies on genetic and environmental issues affecting Zimbabwe populations, with the NBA ensuring regulatory oversight for ethical and safety compliance, thereby enhancing research credibility. Additionally, the NBA's support in securing funding is crucial for advancing impactful research, highlighting the importance its partnership with the University of Zimbabwe. At the UZ farm, the Committee noted the collaboration to integrate biotechnology into agriculture, focusing on food security and productivity through joint research on crop improvement and pest resistance, alongside training programs for farmers and students. Furthermore, the Committee recognised a strategic partnership with Yunnan Industrial Hemp Co., Ltd. to cultivate medicinal cannabis which aims to leverage the global demand for cannabis-derived products. Overally the Committee concluded that these initiatives significantly enhance Zimbabwe's agricultural capabilities and contribute to economic growth and sustainability.

5.0 COMMITTEE OBSERVATION

5.1 The Committee acknowledged the critical roles of National Biotechnology Authority and Verify Engineering in supporting the parent Ministry's Vision 2030, highlighting their importance in advancing the strategic objectives aimed at transforming Zimbabwe into a knowledge-based economy.

5.2 The projects undertaken by both Verify Engineering and NBA are innovation focused and research based and hence the need for Treasury support.

5.3 The Committee also noted that both the Mapfura and Masau projects aligned well with the devolution aspect as espoused by NDS 1 with the emphasis on the participation of the local people in the process. It was noted that the Masau and Mapfuura projects could be springboards for devolution allowing Muzarabani and Mwenezi communities to develop their own niche in the devolution discourse.

5.4 The Committee appreciated the relationship between the locals and NBA in both Muzarabani and Mwenezi, as evidenced by a number of employees and students on work related learning who are employed at the local industry. The internship offered by NBA is in tandem with the vision of the Ministry of producing students who are well versed with knowledge and know how to apply that knowledge.

5.5 The Committee observed that Verify Engineering has several projects which it is currently pursing. It was noted that all these projects were of national importance. However there is need for the entity to focus on one project at a time so that the entity is not overwhelmed by the huge amounts of money required to start different projects.

5.6 The Committee identified a significant opportunity for Verify Engineering to produce and export oxygen, potentially generating foreign currency for Zimbabwe. However, they expressed concern that most of the oxygen is not used in local hospitals because of insufficient storage containers, leading to public hospitals not procuring oxygen from Verify Gases. This underscores a critical gap in market integration with national health institutions

5.7 The Committee noted that Verify Engineering had no official offices and that the infrastructure at Mkwasine Coalfields needs to upgraded if any investor is to take the company seriously.

5.8 The Committee noted that the welfare of staff at both the NBA and Verify Engineering is inadequate, leading to the loss of critical personnel to other countries in the region. Additionally both entities lack proper service vehicles, which further impacts their effectiveness.

5.9 The Committee observed that only one out of all the projects stated by Verify Engineering was fully functional or operational, all the other projects are crippled with the issue of funding.

6.0 RECOMMENDATION

6.1 FOR VERIFY ENGINEERING

6.1.1 The government through the Ministry of Finance and Economic Development should prioritize funding for Verify Engineering projects, particularly those with immediate economic benefits, to facilitate the development of bankable projects .This should be looked into by the first quarter of 2025.

6.1.2 Verify Engineering should prioritize one project at a time to ensure effective resource allocation and successful implementation before moving on to additional initiatives.

6.1.3. Verify Engineering through the Ministry of Higher and Tertiary Education, Innovation, Science and Technology Development should invest in upgrading the infrastructure at Mkwasine Coalfields and other operational sites to attract potential investors. This should be done by 31 March 2025

6.1.4 The Ministry of Industry and Commerce on behalf of Verify Engineering should engage public-private partnership projects like Coal-to-Liquid Fuels and Iron and Steel production. This should be done in the 2nd quarter of 2025.

6.1.5 The Ministry of Health and Child Care should establish local market integration and develop strategies to ensure local hospitals and health institutions procure medical gases produced by Verify Engineering, addressing the current gap in market. This should take place by 25 May 2025.

6.1.6 The Ministry of Higher and Tertiary Education, Innovation, Science and Technology Development should support Verify Engineering with funding to position it as a centre for innovation and research in science and technology, attracting collaborations with local and international research institution. Timeline for this is by 31 August 2025.

6.1.7 Verify Engineering should focus on renewable energy projects to align with global sustainability goals and reduce dependency on non-rewable resources this should be done by 31 August 2025.

6.1.8 Verify Engineering should continue to foster relationships with local communities to ensure project buy-in and support, particulary in resource extraction and industrialisation initiatives. This should be ongoing.

6.2 FOR NATIONAL BIOTECHNOLOGY AUTHORITY

6.2.1 The Ministry of Higher and Tertiary Education, Innovation, Science and Technology Development, in collaboration with the Ministry of Finance Economic Development and Investment Promotion should invest in upgrading the Mapfura processing plant by April 2025to enhance its production capacity for high-value products like wine, oils and juices thereby meeting local and export demand.

6.2.2 National Biotechnology Authority should establish partnerships with local farmers for the sustainable sourcing of Mapfuura and Masau fruits, ensuring a reliable supply chain that benefits the local economy. This should be an ongoing engagement.

6.2.3 National Biotechnology Authority should enhance biosecurity protocols at the Beitbridge Border Post to protect local agriculture from potential pests and diseases while facilitating safe trade of biotechnology products.

6.2.4 National Biotechnology Authority should forge partnerships with local Universities and research institutions to conduct joint research projects on biotechnology applications, driving innovation and knowledge transfer. This should be done by 31 July 2025.

6.2.7 The Ministry of Industry and Commerce together with National Biotechnology Authority should create targeted strategies for exporting products from the Mapfura and Masau projects, identifying potential markets and ensuring compliance with international standards by 31 August 2025.

7.0 CONCLUSION

The inquiry into the operations of Verify Engineering and the National Biotechnology Authority has highlighted their significant potential in contributing to Zimbabwe's industrialisation and economic growth. Both entities play crucial roles in advancing innovation, enhancing local capacities and promoting sustainable practices. However challenges such as funding constrains, infrastructure deficits and the need for enhanced staff welfare have been identified as barriers to their successes. To realise the vision of a competitive industrialised and modernised Zimbabwe by 2030, it is imperative that the government and relevant stakeholders provide the necessary support and resources. By implementing the recommendations outlined in this report, both Verify Engineering and the National Biotechnology Authority can significantly improve their operational efficiencies, foster innovation and ultimately contribute to the country's

development goals. The Committee remains committed to advocating for the interest of these entities and ensuring that their potential is fully realised in the national context.